THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 7 & 8: **Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A constant force of 120 N displaces a body by 4 m in the direction of the force. The work done is: 1

(a) 960 J **(b)** 124 J **(c)** 480 J **(d)** 30 J

2. The kinetic energy of a body of mass 4 kg moving at 5 m s^{-1} is: 1

(a) 20 J **(b)** 50 J **(c)** 25 J **(d)** 100 J

3. The potential energy of a 10 kg body at a height of 3 m (g = 10 m s^{-2}) is: 1

(a) 30 J **(b)** 600 J **(c)** 150 J **(d)** 300 J

4. A machine does 2000 J of work in 8 s. Its power is: 1

(a) 16000 W **(b)** 500 W **(c)** 250 W **(d)** 0 W

5. A ramp of length 10 m raises a load to a height of 2 m (ignore friction). Its mechanical advantage is: 1

(a) 0.2 **(b)** 6 **(c)** 5 **(d)** 10

6. At a point in a medium, 120 density oscillations occur in 6 s. The frequency of the sound wave is: 1

(a) 40 Hz **(b)** 20 Hz **(c)** 0.05 Hz **(d)** 126 Hz

7. A source of frequency 500 Hz produces waves of wavelength 0.4 m in a medium. The speed of the wave is: 1

(a) 400 m s^{-1} **(b)** 1250 m s^{-1} **(c)** 200 m s^{-1} **(d)** 500.4 m s^{-1}

8. A source of frequency 2000 Hz produces waves of wavelength 0.17 m in a medium. The speed of the wave is: 1

(a) 11764.71 m s^{-1} **(b)** 340 m s^{-1} **(c)** 2000.17 m s^{-1} **(d)** 680 m s^{-1}

9. An echo is heard 0.5 s after a shout. If the speed of sound is 340 m s^{-1}, the distance of the reflecting surface is: 1

(a) 42.5 m **(b)** 95 m **(c)** 170 m **(d)** 85 m

10. A sonar signal returns from the ocean floor after 4 s. If the speed in water is 1500 m s^{-1}, the depth is: 1

(a) 3000 m **(b)** 6000 m **(c)** 3010 m **(d)** 1500 m

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Find the potential energy of a 5 kg body at a height of 4 m (g = 10 m s^{-2}). 2

12. State the law of conservation of mechanical energy for a freely falling body. 2

13. Define the amplitude of a sound wave. How is it related to loudness? 2

14. Distinguish between pitch and loudness. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Two bodies of masses m and 4m have the same KE. Find the ratio of their speeds. 3

16. Explain why we hear echoes in a large hall but not in a small room. 3

17. A crane lifts a mass to the 10th floor in time t, then to the 20th floor in 2t. Compare the work and power. 3

OR

A car of mass 1000 kg reaches 72 km h^{-1} from rest in 10 s. Find the power. 3

18. Explain the working of sonar. A signal returns after 0.9 s; find the depth (v = 1530 m s^{-1}). 3

OR

Describe how bats navigate in the dark. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) What are simple machines? 5
(ii) Define mechanical advantage.
(iii) Describe the fixed and movable pulleys.

OR

(i) Describe how an inclined plane makes work easier. 5
(ii) Derive its mechanical advantage.
(iii) A load of 500 N is raised 2 m using a 10 m long smooth ramp. Find the effort.

20. **Case study:** You clap near a cliff and hear an echo after 2 s; v = 340 m s^{-1}. 5
(i) Find the distance to the cliff.
(ii) How far does sound travel in total?
(iii) If the temperature rises, what happens to the time?
(iv) Can you hear a separate echo in a room 5 m wide?

OR

(i) What are ultrasonic waves? 5
(ii) Describe four applications.
(iii) How do bats use them?

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND